COLLEGE OF BUSINESS AND ECONOMICS

SCHOOL OF BUSINESS

BUSINESS INFORMATION TECHNOLOGY

YEAR IV

MODULE TITLE: BUSINESS INTELLIGENCE

LECTURE: RUKUNDO Prince

Analytical Assignment

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REGISTRATION NUMBER: 222020222

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Q1. Does High Revenue Mean High Profit?

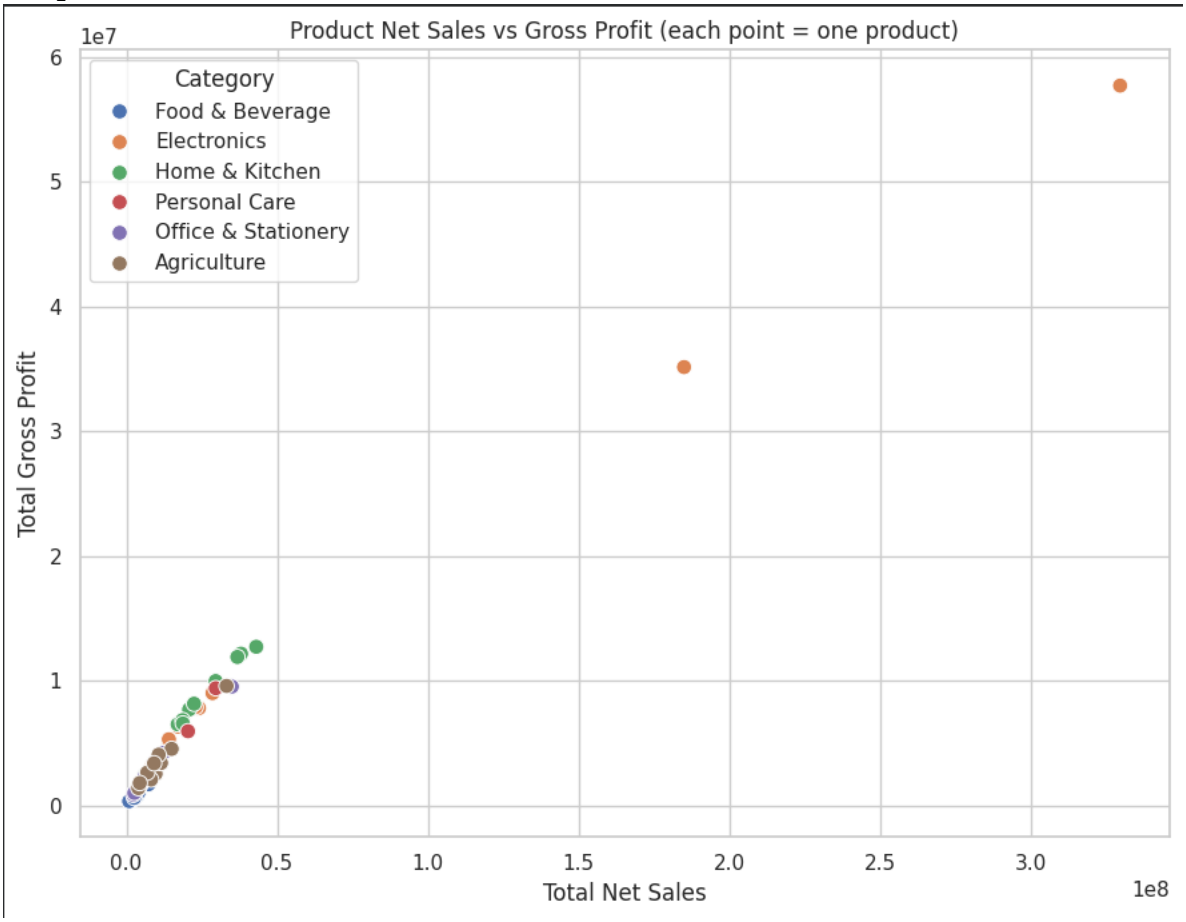
Answer: No.

Expected finding: The product with the highest Net Sales may not have the highest Gross Profit.

Why: Products have different Unit Costs and profit margins.

Two products with similar sales can generate different profits because their costs are different.

Output



Interpretation: At the very top of the list, the assumption largely holds: the two Smartphone products and the leading Home & Kitchen items rank #1–#5 in both sales and profit, so the biggest revenue generators are, in absolute terms, also the biggest profit generators.

However, the assumption breaks down once we move past the top few products. Staple, low-margin items such as Bean Seed 2kg, Maize Seed 2kg (Agriculture) and Rice 5kg, Cooking Oil 1L, Tea 100 Bags (Food & Beverage) sell in decent volumes (mid-table on the sales ranking) but fall several places further down the profit ranking — their profit margins (23%–26%) are noticeably thinner. Conversely, items such as Hand Hoe, Water Bottle, Stapler and Electric

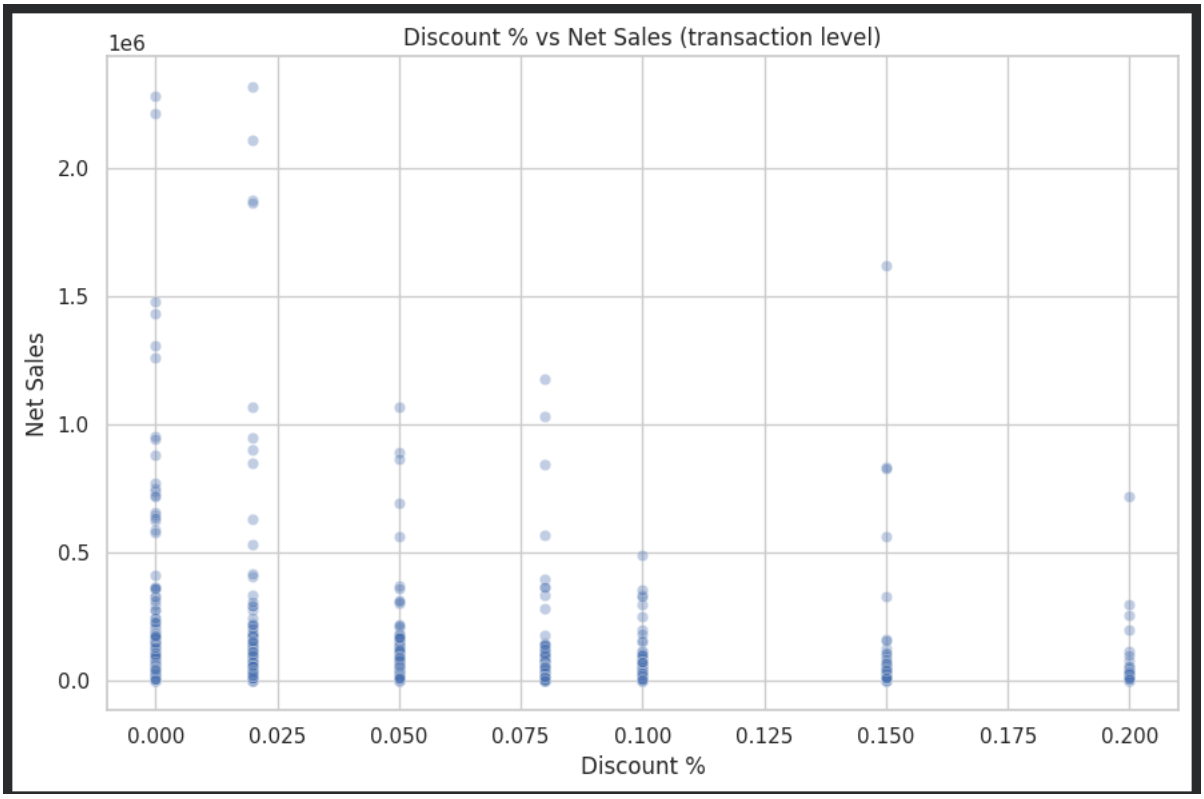
Kettle sell relatively modestly but carry margins of 34%–40%, so they climb several places when ranked by profit instead of sales.

Why can two products with similar sales generate very different profit? Because GrossProfit is driven by the margin (selling price vs. unit cost), not by revenue itself. Two products can generate identical Net Sales while one has a unit cost close to its selling price (thin margin, e.g. electronics where components are expensive) and the other has a unit cost far below its selling price (fat margin, e.g. some home & kitchen or hardware items).

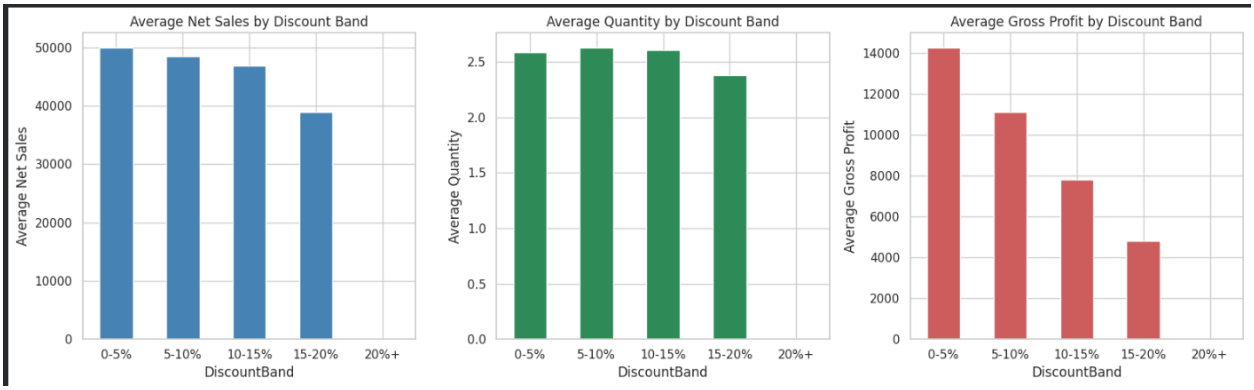
Management implication: Net Sales alone is not a safe way to prioritise products for promotion, shelf space, or marketing spend. Products should be evaluated on both sales volume and profit margin — a high-volume, low-margin staple and a lower-volume, high-margin item can be equally valuable to the business, but for different reasons (footfall/volume vs. profitability).

Q2. Are Discounts Associated With Sales Performance?

Answer: Yes, potentially.
Expected finding: Higher discounts may be associated with higher Quantity or Net Sales, but lower Gross Profit.
Why: Discounts can encourage customers to buy more, while reducing the amount earned per product.
The analysis can show association, not causation.
Output i



Output ii



Interpretation

The correlation between DiscountPct and NetSales is essentially zero (around -0.02), and the correlation with Quantity is also essentially zero (around 0.0004). In other words, transactions with bigger discounts do not, on average, involve buying more units or generating more revenue. The band comparison confirms this: average Net Sales actually falls as the discount band increases (from about 50,000 in the 0–5% band down to about 39,000 in the 15–20% band), average Quantity per transaction stays essentially flat across all bands (~2.4–2.6 units), and average Gross Profit drops sharply as discount increases (from about 14,300 down to about 4,800) — which is expected, since a larger discount directly reduces the revenue collected per unit sold.

What the data allows us to conclude: In this dataset, higher discount levels are associated with lower average Net Sales and substantially lower average Gross Profit per transaction, and are not associated with any increase in units purchased. There is no evidence in this dataset that discounting increases how much customers buy.

What the data does NOT allow us to conclude: We cannot conclude that discounts cause lower sales, or that removing discounts would increase sales. The discount level in this data was not randomly assigned as part of a controlled experiment — it may already reflect other factors we have not controlled for, such as the product category, the customer type receiving negotiated pricing, or the sales channel. This is a classic case of correlation, not causation: a lower Net Sales value on discounted transactions could reflect why a discount was given (e.g. smaller, price-sensitive orders) rather than the discount itself causing the lower value. To test causation properly, management would need a controlled experiment (e.g. a randomised trial offering different discount levels to comparable customers).

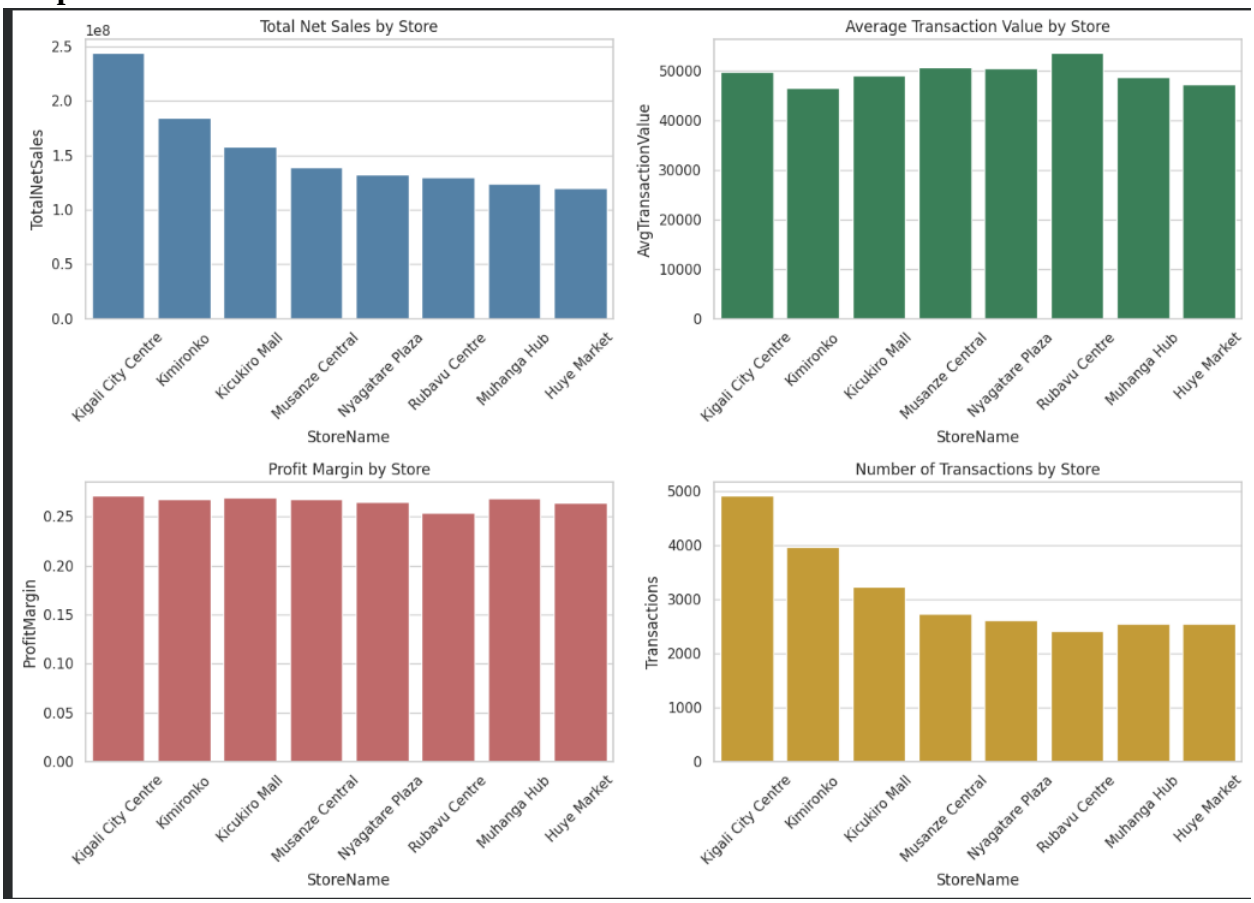
Management implication: the dataset gives no evidence to support the idea that "larger discounts will increase sales" — if anything, larger discounts here are associated with weaker transactions

and clearly erode profit. Before rolling out bigger discounts, management should test the idea in a controlled pilot rather than assume it will work.

Q3. Which Store Is Really Performing Best?

Answer: The best store should be selected using several indicators, not Net Sales alone.
Expected finding: The highest-sales store may not have the highest Gross Profit or Profit Margin.
Why: Stores can have different transaction volumes, quantities, and profit margins.
Key measures: Net Sales, Gross Profit, Profit Margin, Average Transaction Value, and transactions.

Output



Interpretation

Kigali City Centre leads on Net Sales, transaction count, units sold and gross profit, and also has the highest profit margin (about 27.2%) among all eight stores — so in this dataset it is genuinely the strongest store overall, not just the biggest by revenue.

But the ranking is not consistent across every measure for the other stores, which is exactly the warning the assignment asks us to investigate. Rubavu Centre ranks only 6th by Total Net Sales, yet it has the 2nd-highest Average Transaction Value of all stores (≈53,500, second only overall)

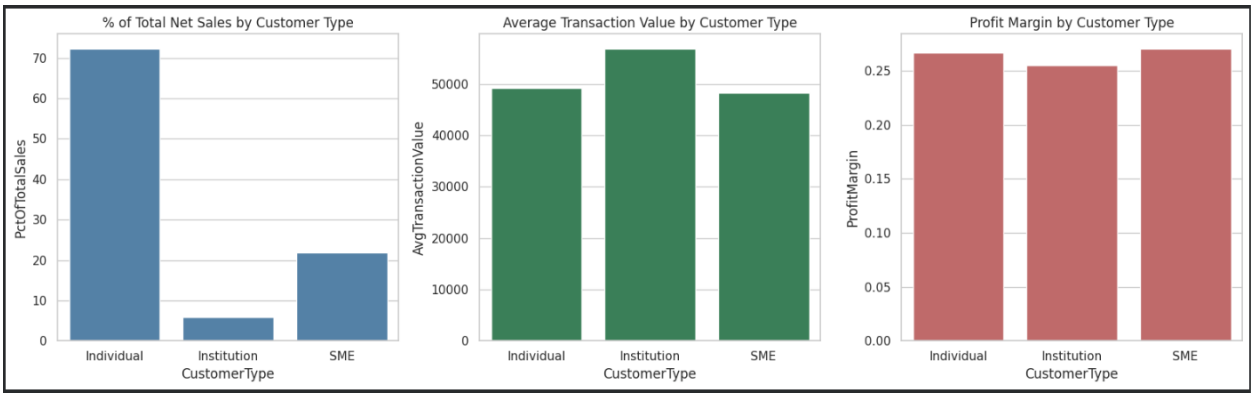
— its customers buy in relatively large baskets. Despite this, Rubavu Centre has the lowest profit margin of all 8 stores (≈25.4%, versus 26.4%–27.2% elsewhere). A manager looking only at "average transaction value" or "Net Sales" would rate Rubavu Centre as a healthy, high-value store, while in fact it is converting that revenue into profit less efficiently than every other store.

Why might the highest-Net-Sales store not be the strongest store overall? Net Sales combines two very different things — how many transactions a store processes and how large and profitable each transaction is — into a single number. A store can reach high Net Sales through sheer transaction volume while carrying a lower margin, or through fewer but larger transactions that still erode profit if the cost base is higher (e.g. different product mix, more discounting, different customer types). Only by looking at Transactions, Average Transaction Value and Profit Margin side by side can management see which stores are efficient, which are large-but-thin-margin, and which genuinely combine scale with profitability.

Q4. Which Customer Type Creates the Most Business Value?

Answer: The customer type with the strongest overall combination of sales and profitability.
Expected finding: The highest-sales customer type may not be the most profitable.
Why: Individual, Institution, and SME customers may differ in transaction size, quantity, and profit margin.
The analysis should compare all three types before selecting the strongest one.

Output



Interpretation

Individual customers dominate the business: 3,600 of the roughly 4,975 customers (about 72%), about 18,100 transactions, and about 72% of total Net Sales. SMEs contribute a solid second place (about 22% of sales from only 1,119 customers), while Institutions are a small group (256 customers) but contribute meaningfully (≈6% of sales) with by far the highest average transaction value (≈56,900 vs. ≈48,000–49,000 for the other two types) — consistent with institutions placing fewer, larger, bulk orders.

On profitability, the picture changes: SME customers have the highest profit margin (≈27.1%), Individuals sit in the middle (≈26.7%), and Institutions have the lowest margin (≈25.5%) despite their large average order size.

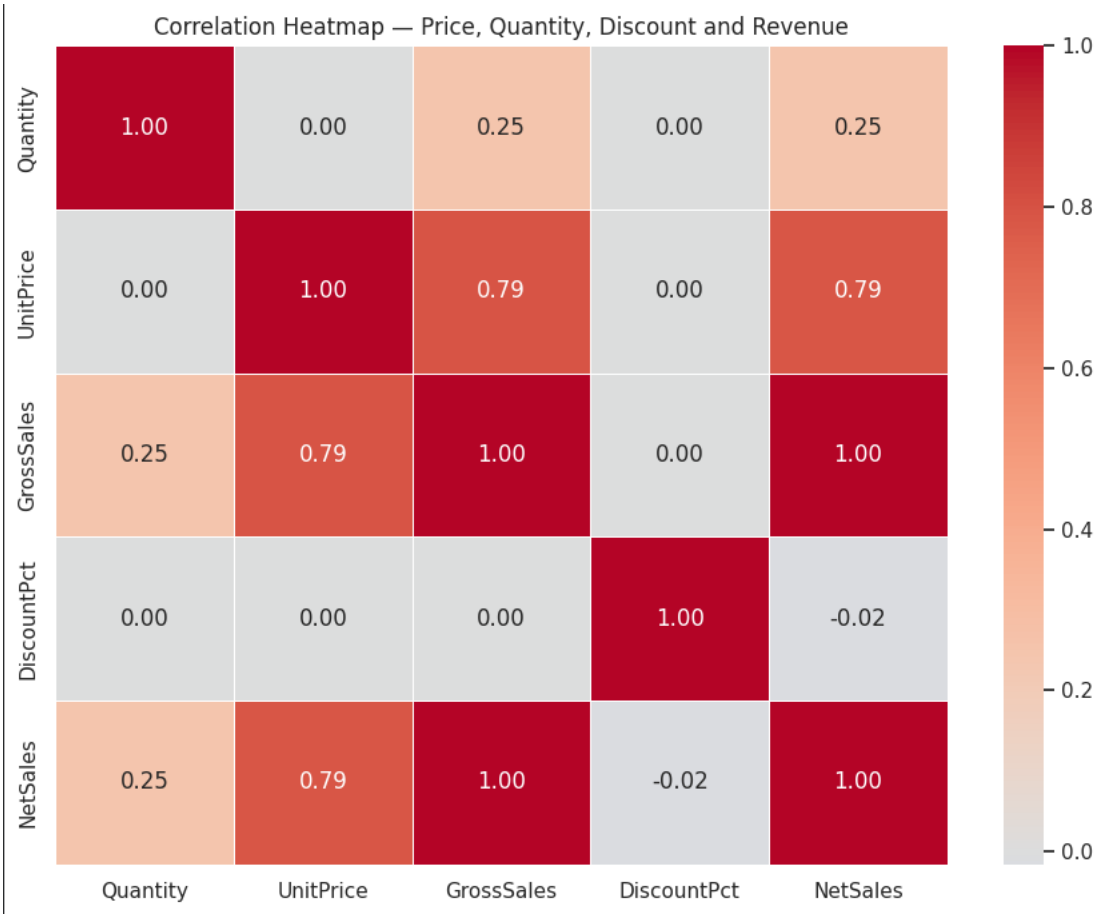
Is the customer type with the highest sales also the most profitable? No. Individuals generate by far the most sales (72%) but are not the most profitable customer type on a margin basis — SMEs are. Institutions generate large average transactions but convert them into profit least efficiently, likely because bulk/negotiated pricing compresses their margin.

Management implication: the business should not treat "customer type with the biggest revenue share" and "customer type management should prioritise for profitability" as the same thing. Individuals remain the core volume driver and should not be neglected, but SMEs deserve attention as the most profit-efficient segment, and any further discounting or bulk-pricing policy for Institutional accounts should be reviewed carefully, since it is already the lowest-margin segment.

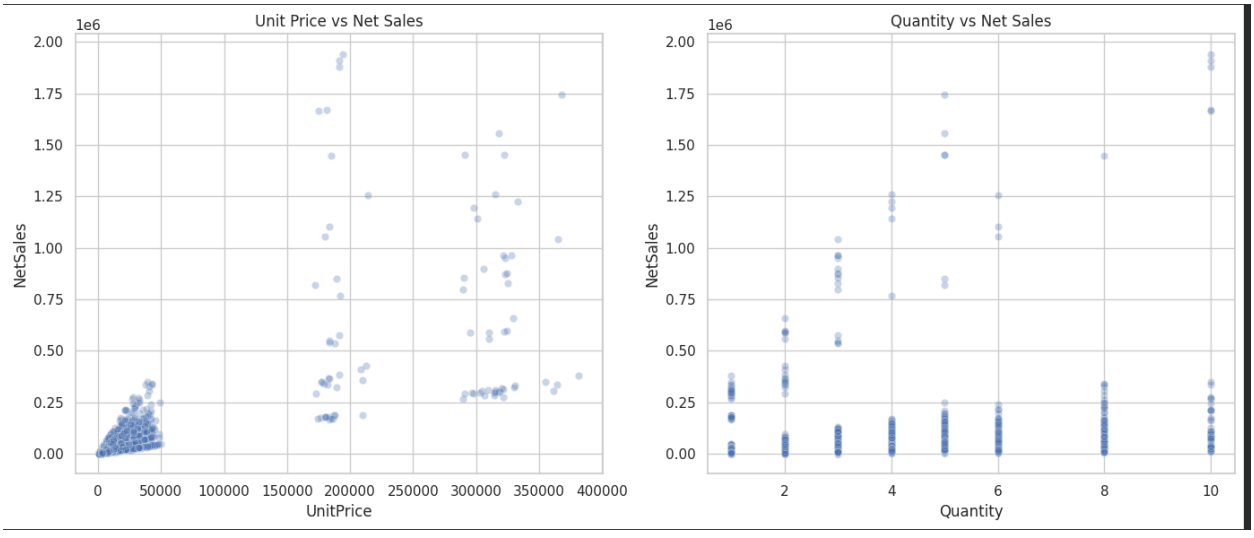
Q5. What Is the Relationship Between Product Price, Quantity and Revenue?

> **Answer**: Strong relationships are expected between Quantity, UnitPrice, and NetSales.
> Expected finding: Quantity and UnitPrice will likely have strong relationships with NetSales.
> Why: Sales value is mathematically based on price and quantity.
> Therefore, some correlations are expected from the formula and are not necessarily new business insights.

Output i



Output ii



Interpretation

NetSales correlates almost perfectly with GrossSales (≈0.999). UnitPrice correlates strongly with NetSales (≈0.79), while Quantity correlates only moderately (≈0.25), and DiscountPct correlates essentially not at all with NetSales (≈-0.02).

Which of these are genuine business insights, and which are data-generated artefacts of the formulas?

GrossSales ↔ NetSales (≈0.999) and UnitPrice ↔ GrossSales (≈0.79) are not real discoveries — GrossSales = Quantity × UnitPrice and NetSales ≈ GrossSales × (1 − DiscountPct) by construction. A near-perfect correlation here simply confirms the formula was applied correctly; it tells management nothing new about customer behaviour.

The comparison between UnitPrice (≈0.79) and Quantity (≈0.25) as drivers of NetSales is a genuine, useful business insight: transaction value in this business is driven far more by which, higher-priced product is bought than by how many units are bought in one visit. This tells management that revenue growth is more likely to come from selling higher-value products (upselling, product mix) than from encouraging customers to buy more units of the same item.

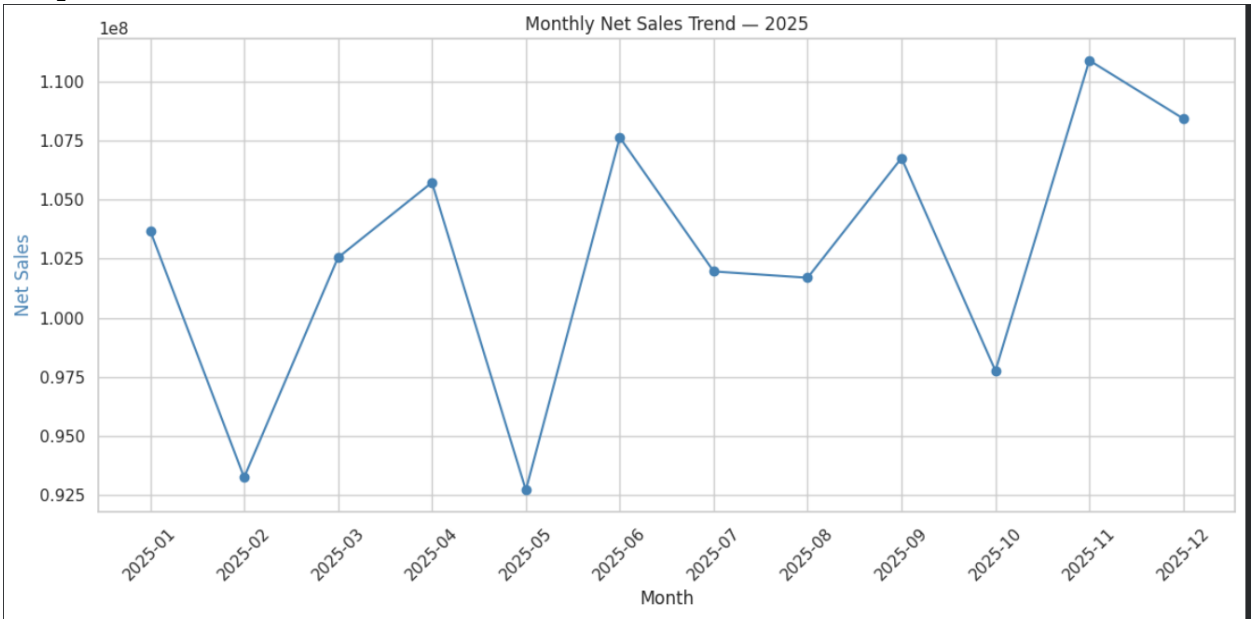
The very weak DiscountPct ↔ NetSales correlation (≈-0.02) reinforces the Question 2 finding: in this dataset, discounting is not a meaningful lever for either raising or lowering the size of a transaction once price and quantity are already accounted for.

Management implication: focus merchandising and sales effort on product mix and pricing (which move the needle), rather than assuming basket size (Quantity) or discounting are the main levers of revenue — neither shows a strong relationship with NetSales in this dataset.

Q6. Does the Company Have a Monthly Sales Pattern?

> **Answer**: Yes, if the data shows consistent changes over time.
> Expected finding: Some months will have higher or lower Net Sales than others.
> Why: Customer demand, transaction numbers, and quantities sold can change from month to month.
> A repeated pattern at similar times would provide stronger evidence of seasonality.

Output



Interpretation

Net Sales moves up and down noticeably from month to month (roughly between 93 million and 111 million), with no single month standing far apart from the rest. Quarterly totals show a mild, fairly steady upward drift across the year (Q1 lowest, Q4 highest, an increase of roughly 6% from Q1 to Q4), rather than one dramatic jump.

What evidence suggests performance changes over time? Two things: (1) the month-to-month percentage changes swing by double digits in both directions (the largest single-month jump and the largest single-month drop are both more than 10%), showing real volatility rather than a flat line; and (2) the quarter-by-quarter totals rise consistently from Q1 through Q4, which is a genuine, measurable trend within this year's data.

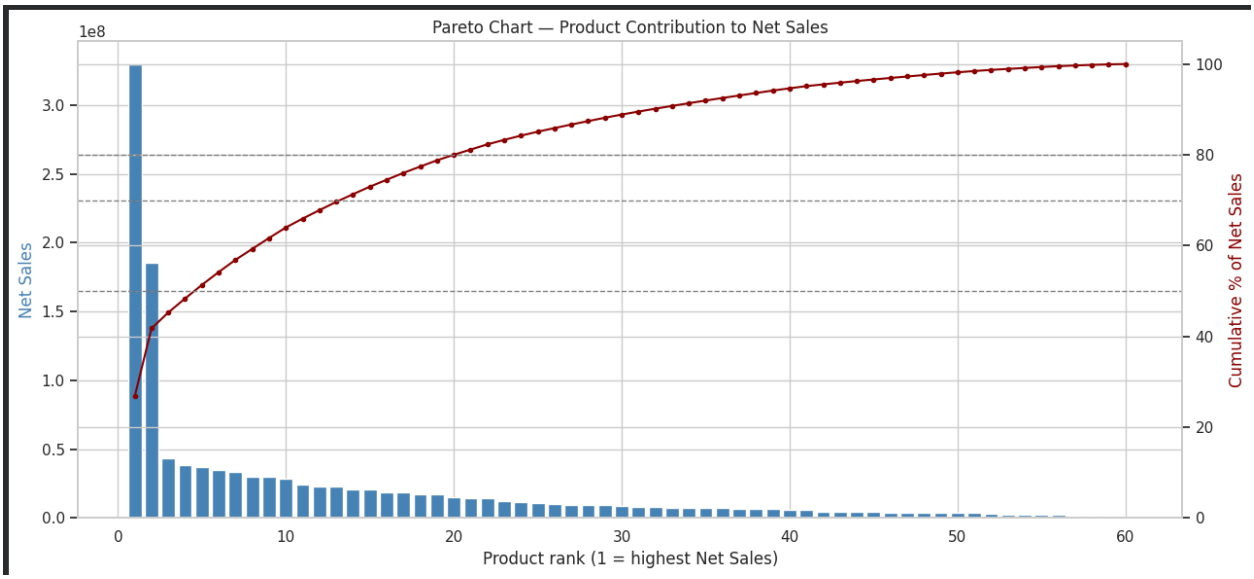
Why we do not call this "seasonality" yet: seasonality means a pattern that repeats in the same calendar months every year (e.g. always higher in December). With only twelve months of data from a single year, we cannot distinguish a true recurring seasonal effect from a one-off trend, random month-to-month noise, or a business-specific event (a promotion, a new store opening, stock availability) that happened to occur in a particular month. Confirming seasonality would require at least two or three years of comparable monthly data showing the same months consistently outperforming or underperforming.

Management implication: treat the observed monthly swings as evidence that performance is not flat, and keep tracking month-to-month change as a KPI, but avoid labelling any specific month as "seasonally strong/weak" or building inventory and staffing plans around an assumed yearly cycle until more years of data confirm the pattern repeats.

Q7. Do a Small Number of Products Generate Most Revenue?

Answer: Yes, if the cumulative analysis shows high concentration.
Expected finding: A relatively small number of products may generate a large percentage of total sales.
Why: Some products may have much higher demand than others.
The Pareto analysis will show how many products generate 50%, 70%, and 80% of revenue.

Output



Interpretation

Revenue is highly concentrated: only about 5 of the 60 products (roughly 8%) are needed to reach 50% of total Net Sales, about 14 products (roughly 23%) reach 70%, and about 21 products (roughly 35%) reach 80%. In other words, roughly one-third of the catalogue generates four-fifths of total revenue, while the remaining two-thirds of products together contribute only about a fifth of sales. This concentration is largely driven by a small number of high-value Electronics items (notably the two smartphone products), which alone make up a very large share of total revenue.

Is revenue highly concentrated among a small group of products? Yes.

Implications for inventory and product management:

The small set of "star" products deserves priority in stock availability — running out of a top-5 product has an outsized impact on total revenue, so safety-stock levels and reorder frequency for these items should be higher than for the long tail.

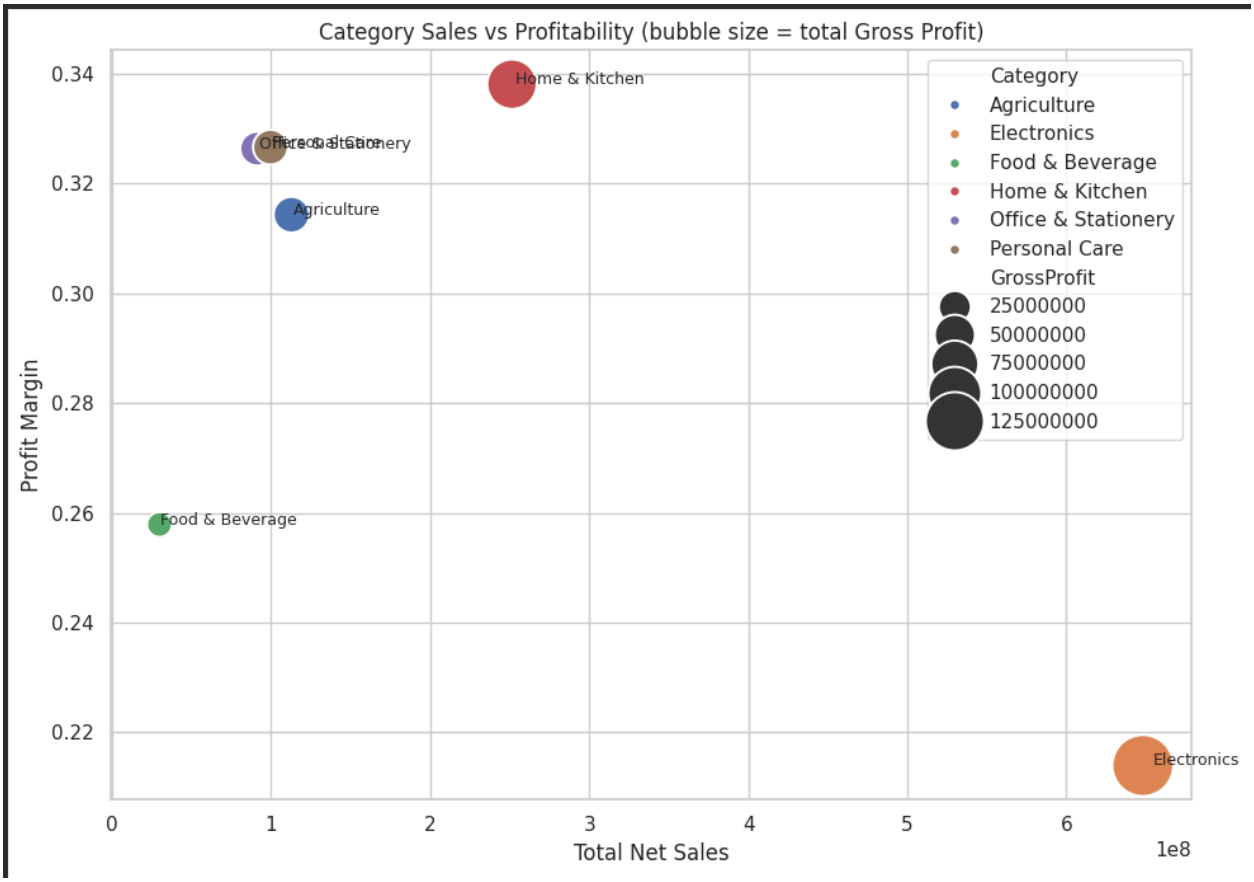
The large "long tail" of low-contribution products should be reviewed for rationalisation: some may be candidates for reduced shelf space, bundling, or discontinuation if they tie up inventory capital and staff time without contributing meaningfully to revenue — although Question 1's findings show some of these lower-revenue items carry strong margins, so profitability (not just sales) should be checked before removing any product.

Because revenue depends so heavily on a handful of SKUs, the business also carries concentration risk — a supply disruption or a competitor undercutting price on just one or two top products could materially affect total company revenue.

Q8. Which Product Categories Are Driving the Business?

Answer: The categories with the highest combination of Net Sales and Gross Profit.
Expected finding: Some categories will contribute more sales and profit than others.
Why: Categories differ in demand, prices, costs, and product mix.
A category can have high sales but a relatively low margin, so both sales and profitability must be considered.

Output



Interpretation

Electronics — high sales, relatively low margin. Electronics contributes about 53% of total Net Sales, by far the largest share, but its profit margin (≈21%) is the lowest of all six categories — a direct echo of Question 1's finding about the two smartphone products.

Home & Kitchen — high sales AND high margin. Home & Kitchen is the second-biggest category (≈20% of sales) and also carries the highest margin of all categories (≈34%) — currently the strongest combination of scale and profitability in the business.

Agriculture, Personal Care, Office & Stationery — lower sales, relatively high margin. Each contributes only about 7–9% of total sales individually, but all three carry healthy margins in the 31%–33% range.

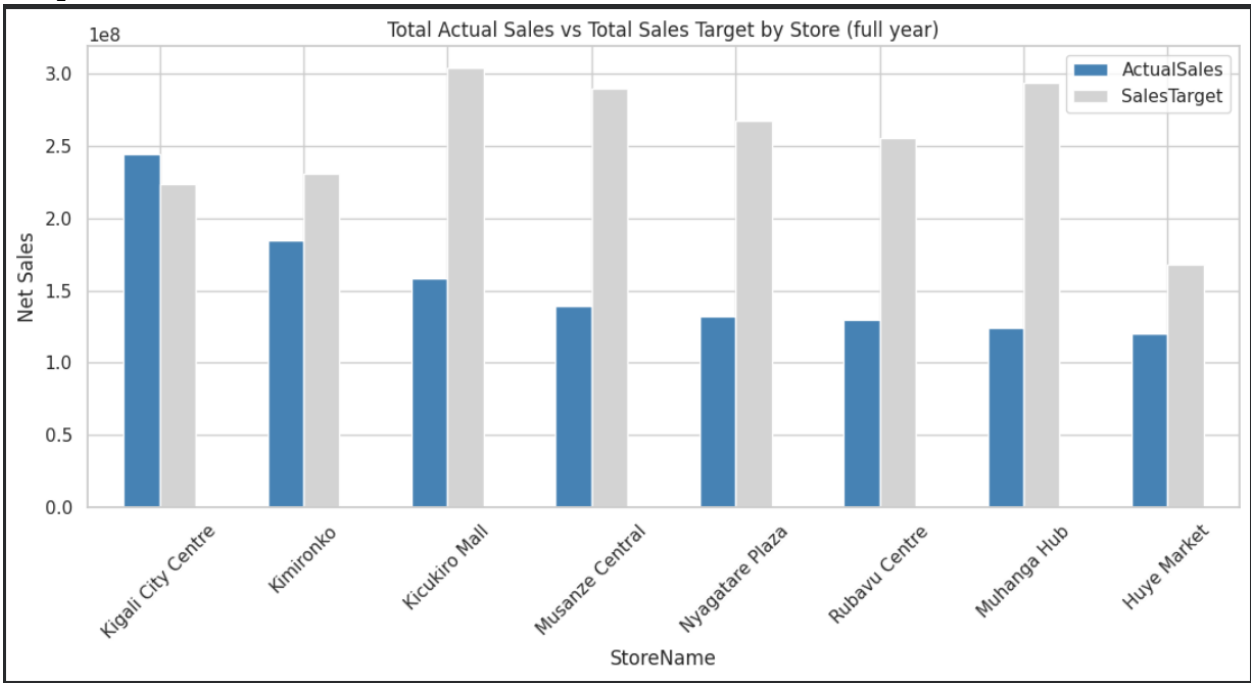
Food & Beverage — lowest sales, and a below-average margin. This category contributes only about 2.4% of total sales, the smallest of all six, and its margin (≈26%) is also below every category except Electronics — it is currently the weakest overall performer on both dimensions.

What management should investigate further: (1) why Electronics carries such a thin margin relative to its scale — is it competitive pricing pressure, high component costs, or heavy discounting on these specific items — and whether the margin can be improved without losing the volume; (2) whether Food & Beverage's low revenue reflects genuinely low customer demand, insufficient shelf space/assortment, or pricing that is too low relative to cost, since it is underperforming on both sales and margin simultaneously; and (3) whether Home & Kitchen's strong combination of sales and margin can be replicated by promoting it further, or by applying lessons from its pricing/cost structure to other categories.

Q9. Can the Company Meet Sales Targets?

Answer: Yes, for some store-months; No, for others.
Expected finding: Some stores will exceed their targets while others will fall below them.
Why: Actual sales differ according to customer demand, transaction volume, and store performance.
Comparing Actual Sales with Targets shows whether planned performance was achieved.

Output



Interpretation

Only 9 of the 96 store-month combinations (≈9%) actually met or exceeded their target for the year — and every single one of those 9 belongs to just two stores: Kigali City Centre (8 months) and Kimironko (1 month), both urban Kigali locations. Every other store-month across the remaining six stores, for all twelve months, missed its target.

Looking at overall achievement: Kigali City Centre is the only store with an average achievement above 100% (≈109%) and the only store with a positive total variance for the year (actual sales roughly 21 million above target, cumulatively). At the other end, Muhanga Hub has both the weakest average achievement (≈42%) and the largest total negative variance (about −170 million) for the year.

Store-months that exceeded target: all 8 at Kigali City Centre, plus 1 at Kimironko (March).

Store-months that failed to reach target: all remaining 87 combinations (6 of the 8 stores missed their target in every single month of the year).

Strongest overall target achievement: Kigali City Centre.

Largest total positive variance: Kigali City Centre.

Largest total negative variance: Muhanga Hub.

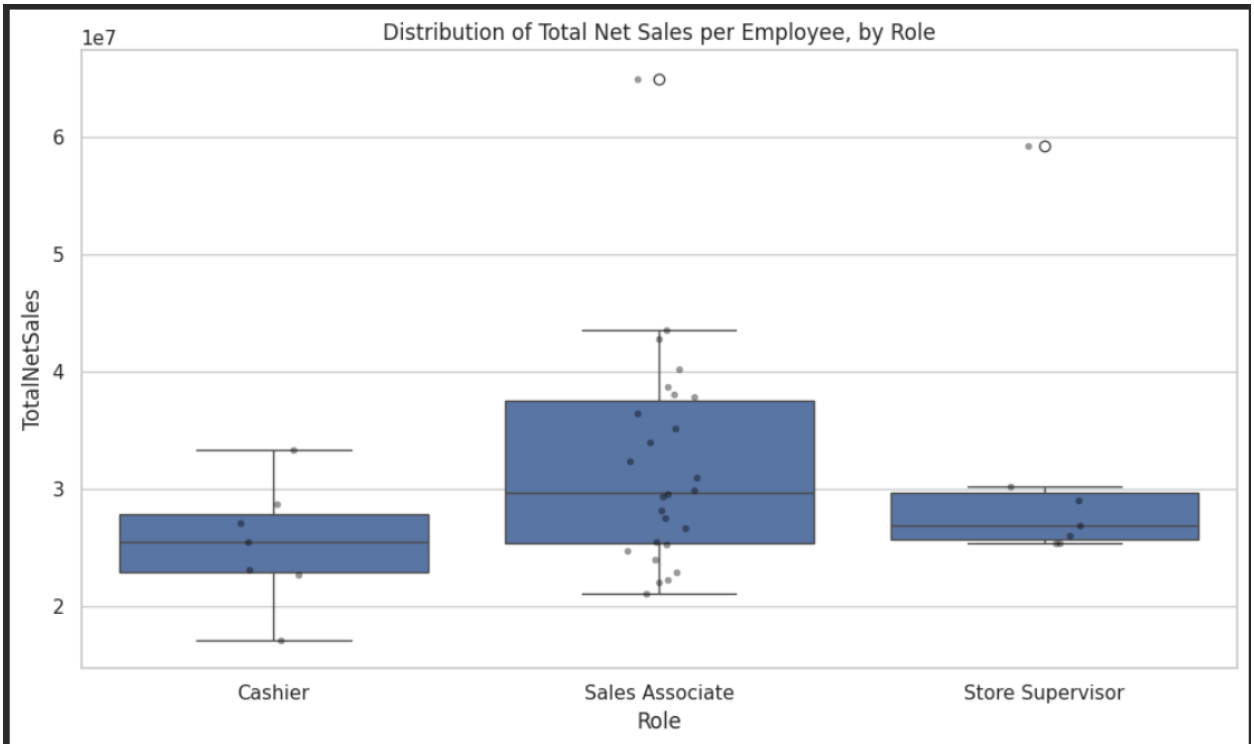
Why does target achievement tell management something raw sales figures do not? Raw Net Sales ranks stores purely by size — and by that measure, Kigali City Centre also happens to come out on top, so in this case the two views agree at the very top. But raw sales says nothing

about whether a store is meeting the expectations set for it. Target achievement reveals a much more urgent story that Net Sales alone hides completely: six of the eight stores are missing target every single month of the year, several by a wide margin (down to 30%–40% achievement in their worst months). A manager looking only at the Net Sales ranking would see "8 stores, roughly ordered from strong to weaker" and might conclude performance is broadly fine; the target-achievement view instead shows a company-wide shortfall concentrated outside the two Kigali city stores, which calls for investigating whether targets for the secondary-city stores were set unrealistically, or whether those stores genuinely need operational support.

Q10. Employee Performance and Management Decision

Answer: No, sales alone cannot determine the best employee.
Expected finding: The employee with the highest Net Sales may not have the highest Gross Profit or Average Transaction Value.
Why: Employees have different roles, transaction volumes, and store assignments. Performance should therefore be compared using sales, transactions, quantity, Gross Profit, and Profit Margin.

Output



Interpretation

There are observable differences between employees and between roles, but they line up closely with store assignment and role, not obviously with individual skill:

Store Supervisors handle the most transactions on average (≈653), followed by Sales Associates (≈631) and Cashiers (≈573) — yet Cashiers have the highest average profit margin (≈28.4%), ahead of Sales Associates (≈26.5%) and Store Supervisors (≈26.2%). Handling more transactions does not translate into the highest margin.

The employees with the highest total Net Sales, the highest average transaction value, and the highest Gross Profit are three different people in three different roles and stores — no single employee dominates on every measure, and the top performers by total sales work at the busiest store (Kigali City Centre / Store 8, Muhanga Hub), which mechanically gives them more transactions to work with.

The scatter of Transactions vs Total Net Sales shows total sales rising with the number of transactions handled, which is expected — an employee working more transactions will tend to generate more total sales almost by definition, regardless of individual skill.

Why should management be cautious about interpreting employee-level performance using sales alone?

Store assignment confounds the comparison. Employees at a busier store (more footfall, more transactions available to handle) will naturally rack up higher total Net Sales than an equally skilled employee at a quieter store — the dataset cannot separate "store traffic" from "individual selling skill."

Role changes what "high performance" should even look like. Store Supervisors may spend time on supervisory duties not captured as sales transactions, so comparing their total Net Sales directly against Sales Associates or Cashiers, who spend all their time processing transactions, is not a fair comparison of individual sales ability.

Volume and margin can move in opposite directions. The role with the most transactions per employee (Store Supervisors) has the lowest average margin among the three roles, while the role with fewer transactions (Cashiers) has the highest margin — total sales alone would credit the wrong role/employee with the "best" performance if margin is also part of what management cares about.

Management implication: before rewarding or penalising individual employees, performance should be benchmarked within the same role and, ideally, the same store, and should combine transaction volume, average transaction value, and profit margin rather than total Net Sales in isolation.